FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent 8 lessons investigating the "Shrinking Uji Cup" phenomenon — discovering why doubling every dimension of a container makes it hold eight times as much, not two times. Now it is your turn to bring all of your learning together into one well-reasoned, evidence-based explanation.

READ THE DRIVING QUESTION CAREFULLY:
"Why does doubling the size of a container make it hold eight times as much — and how can we use this mathematical pattern to design, build, and predict the properties of any similar shape or object?"

YOUR TASK:
Answer the driving question fully by completing each section below. In every section, connect your mathematical reasoning back to the uji cups from the class demonstration, AND apply your understanding to a new real-life Kenyan context. Use diagrams, calculations, or tables wherever they strengthen your explanation.

TIPS FOR SUCCESS:

- Show all working — a correct answer without reasoning earns fewer marks.
- Use precise mathematical vocabulary: scale factor, linear scale factor (k), area scale factor (k^2), volume scale factor (k^3), similar figures, enlargement, ratio, proportion.
- Make your explanation clear enough that a fellow Grade 10 student who missed the lessons could read it and understand the big idea.
- Check your calculations at least once before submitting.

Section 1: The Mystery of the Uji Cups — Identifying the Pattern	
Describe what was observed with the three uji cups. Explain what a linear scale factor (k) is, and use measurements or a table to show how the height	When our class measured the three uji cups — the toddler cup, the adult mug, and the catering flask — we found that every dimension of the adult mug was exactly twice the corresponding dimension of the toddler cup, and every dimension of the catering flask was exactly three times the toddler cup's dimensions. This multiplying value is called the linear scale factor, k. For the mug, k = 2; for the flask, k = 3.

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<td>and diameter of the cups are related. Then state — in your own words — why filling the larger cup required more water than most people expected. What is the mathematical pattern at the heart of this mystery?</td>
<td>A table of measurements might look like this: | Cup	Height (cm)	Diameter (cm)	k (compared to toddler cup)
Toddler	6	4	1
Adult mug	12	8	2
Catering flask	18	12	3 | Most people expected the adult mug to hold twice as much water as the toddler cup, because 'twice as big' sounds like 'twice as much.' But this thinking only applies to one dimension (length). A cylinder has THREE dimensions that all scale at the same time. The cross-sectional area (a circle) scales by $k^2 = 4$, and the volume scales by $k^3 = 8$. So the adult mug holds 4 times as much water as the toddler cup, and the catering flask holds 27 times as much — not 3 times. The mathematical pattern is: when you scale every linear dimension by k, areas scale by k^2 and volumes scale by k^3.</td>
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<th colspan="2">Section 2: The Mathematics of Scaling — Proving Why k, k², and k³ Work</th>
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<td>Using the formulae for a cylinder (or any shape of your choice), prove algebraically why scaling every length by a factor of k causes area to scale by k^2 and volume to scale by k^3. Show full working. Then verify your proof using the actual numbers from the uji cup demonstration (k = 2 and k = 3).</td>
<td>Let the toddler cup have radius r and height h.

Its cross-sectional area is: $A_1 = \pi r^2$
Its curved surface area is: $CSA_1 = 2\pi rh$
Its volume is: $V_1 = \pi r^2 h$

Now scale every linear dimension by k. The new cup has radius kr and height kh.

New cross-sectional area: $A_2 = \pi(kr)^2 = \pi k^2 r^2 = k^2 \times A_1$ ✓
New curved surface area: $CSA_2 = 2\pi(kr)(kh) = 2\pi k^2 rh = k^2 \times CSA_1$ ✓
New volume: $V_2 = \pi(kr)^2(kh) = \pi k^2 r^2 \times kh = \pi k^3 r^2 h = k^3 \times V_1$ ✓

This algebraic proof works for ANY shape, not just cylinders, because every area formula contains exactly two linear dimensions multiplied together (giving k^2), and every volume formula contains exactly three linear dimensions multiplied together (giving k^3).

Verification with the uji cups:
– For k = 2 (toddler → adult mug): Area scales by $2^2 = 4$ ✓ (we needed 4 cups of water to fill the cross-section), Volume scales by $2^3 = 8$ ✓
– For k = 3 (toddler → catering flask): Area scales by $3^2 = 9$ ✓ (the label requires 9 times as much paper), Volume scales by $3^3 = 27$ ✓

The algebra and the physical experiment agree perfectly, which gives us strong confidence in the rule.</td>
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<th>Section 3: Applying the Pattern — Similarity in a Kenyan Context</th>
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Choose ONE real-life Kenyan context (for example: a farmer in Kericho designing water storage tanks of different sizes; a jua kali artisan in Kamukunji making sufurias of similar shapes; a baker in Mombasa scaling up a mandazi recipe using similar baking tins; or an architect in Nairobi scaling a building model). Describe the context, identify the linear scale factor, and then calculate at least TWO of the following: the new surface area, the new volume, or the new capacity. Show all working and explain what your answer means in practical terms.	Context: A jua kali artisan in Kamukunji, Nairobi makes aluminium sufurias (cooking pots) in similar cylindrical shapes. The small sufuria has a radius of 10 cm and a height of 12 cm. The large family sufuria is similar to the small one, with a linear scale factor of $k = 2.5$. Step 1 – Find the dimensions of the large sufuria: Radius: $10 \times 2.5 = 25$ cm Height: $12 \times 2.5 = 30$ cm Step 2 – Calculate the surface area scale factor: Area scale factor = $k^2 = 2.5^2 = 6.25$ Small sufuria total surface area = $2\pi r(r + h) = 2\pi(10)(10 + 12) = 2\pi(10)(22) = 440\pi \approx 1{,}382$ cm² Large sufuria total surface area = $6.25 \times 440\pi = 2{,}750\pi \approx 8{,}639$ cm² Practical meaning: The artisan needs 6.25 times as much aluminium sheet to make the large sufuria. This helps him price the larger pot correctly — it costs far more than 2.5 times the small one. Step 3 – Calculate the volume scale factor: Volume scale factor = $k^3 = 2.5^3 = 15.625$ Small sufuria volume = $\pi r^2 h = \pi(10)^2(12) = 1{,}200\pi \approx 3{,}770$ cm³ ≈ 3.77 litres Large sufuria volume = $15.625 \times 1{,}200\pi = 18{,}750\pi \approx 58{,}905$ cm³ ≈ 58.9 litres Practical meaning: The large sufuria holds about 15.6 times as much food as the small one — enough to cook githeri for a large community gathering, not just a small family.

Section 4: Working Backwards — Finding the Scale Factor from Area or Volume

Sometimes you know the area ratio or volume ratio, but you need to find the linear scale factor. Explain how to work backwards from k^2 or k^3 to find k. Then solve this problem fully: A water tank manufacturer in Kisumu produces two similar tanks. The smaller tank has a capacity of 500 litres. The larger tank has a capacity of 4,000 litres. (a) Find the volume scale factor. (b) Find the linear scale factor, k. (c) If the smaller tank has a total surface area of 3.2 m², find the surface area of the larger tank.	To work backwards from a volume ratio to find k, you take the cube root: $k = \sqrt[3]{\text{volume scale factor}}$. To work backwards from an area ratio to find k, you take the square root: $k = \sqrt{\text{area scale factor}}$. This is the reverse of what we proved in Section 2. Solution: (a) Volume scale factor = larger volume ÷ smaller volume = 4,000 ÷ 500 = 8 (b) Linear scale factor: $k^3 = 8$, so $k = \sqrt[3]{8} = 2$ The larger tank has every linear dimension exactly twice that of the smaller tank. (c) Area scale factor = $k^2 = 2^2 = 4$ Surface area of larger tank = 4×3.2 m² = 12.8 m² Interpretation: The manufacturer in Kisumu needs 4 times as much material (metal sheeting or plastic) to build the large tank as the small one, even though the large tank holds 8 times as much water. This is an important business insight — material costs scale by

	k^2, but storage capacity scales by k^3, so larger tanks are more cost-efficient per litre of water stored. This is why big water storage projects (like those supplying Kisumu or Nairobi estates) use the largest tanks possible.

Section 5: Connecting Back to the Driving Question — Your Final Model	
Answer the driving question directly and completely, using everything you have learned across all 8 lessons. Your answer must: (1) Explain WHY doubling every dimension leads to 8 times the volume (not 2 times). (2) State the general rule for any scale factor k. (3) Explain how this pattern can be used to DESIGN, BUILD, and PREDICT properties of similar shapes. (4) Reflect on one thing this unit changed about how you see the world around you in Kenya.	Doubling the size of a container makes it hold eight times as much — not two times — because volume depends on THREE dimensions (length × width × height, or radius² × height). When you double every single dimension, each of the three dimensions is multiplied by 2, so the total volume is multiplied by $2 \times 2 \times 2 = 2^3 = 8$. You cannot just double the volume by doubling one measurement; all dimensions scale together. The General Rule: For any two similar shapes or solids with linear scale factor k: – Corresponding lengths scale by k (e.g., all sides, heights, radii, perimeters) – Corresponding areas scale by k^2 (e.g., cross-sections, surface areas, footprints) – Corresponding volumes (and capacities) scale by k^3 Using the Pattern to Design, Build, and Predict: Design: An architect in Nairobi designing a scale model of a building at 1:50 scale knows that every real dimension is 50 times the model dimension. The real building's floor area is $50^2 = 2{,}500$ times the model's floor area, and its volume is $50^3 = 125{,}000$ times the model's volume. She can predict these values without ever leaving her office. Build: A jua kali artisan making similar-shaped pots of different sizes can calculate exactly how much aluminium sheet is needed for each size (using k^2) and how much each pot will hold (using k^3), reducing waste and improving pricing. Predict: Engineers designing water tanks, fuel storage containers, or grain silos across Kenya can use k^3 to predict capacity and k^2 to estimate material costs — making projects cheaper and more efficient. Personal Reflection: Before this unit, I assumed that if something looked 'twice as big,' it held twice as much. Now I realise that our everyday language about size is misleading — 'twice as big' usually means twice as tall or twice as wide, but the volume grows much faster than that. When I walk through a market in Nairobi and see large and small versions of the same container, I now automatically think: what is k? And then k^2 and k^3 follow immediately. This unit has genuinely changed the way I see and question the physical world.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**

Understanding of Scale Factor (k, k^2, k^3)	Clearly and correctly defines linear scale factor (k), area scale factor (k^2), and volume scale factor (k^3). Accurately applies all three in every section. Consistently connects the three scaling rules back to the uji cup phenomenon with precise values.	Correctly defines and applies k, k^2, and k^3 in most sections. Makes the connection to the uji cups but may miss one specific value or detail. Minor errors do not affect the overall understanding shown.	Shows partial understanding of k but confuses k^2 and k^3, or applies only one scaling rule consistently. The connection to the uji cup phenomenon is vague or incomplete. Significant errors in at least two sections.
Algebraic Proof and Mathematical Reasoning	Provides a complete, accurate algebraic proof showing why areas scale by k^2 and volumes scale by k^3 for a chosen shape. All working is clearly shown and logically ordered. Verification using $k = 2$ and $k = 3$ from the uji cups is accurate and explicitly stated.	Provides a mostly correct algebraic proof with clear working. May have a minor arithmetic or notation error that does not undermine the logic. Verification with the uji cup values is attempted and substantially correct.	Attempts an algebraic proof but makes significant errors in substitution or simplification, OR provides only numerical examples without a general algebraic argument. Working is incomplete or difficult to follow.
Application to a Kenyan Real-Life Context	Selects a specific, well-described Kenyan context (named location, realistic values). Correctly identifies k and calculates at least two of: surface area, volume, or capacity, with full working shown. Explains the practical meaning of each answer clearly and specifically.	Selects a relevant Kenyan context and correctly calculates at least one of the required values with working shown. The practical meaning is stated but may lack specific detail. Minor calculation errors are present but do not indicate a conceptual misunderstanding.	Selects a context but it lacks Kenyan specificity or realistic values. Only one calculation is attempted, or calculations contain major errors. The practical meaning of results is not explained or is incorrect.
Working Backwards (Finding k from k^2 or k^3)	Clearly explains the method for finding k from a given area or volume ratio (square root and cube root respectively). Correctly solves all three parts of the Kisumu water tank problem with full working. Provides a meaningful interpretation of the answers in context.	Explains the reverse method adequately and solves at least two of the three parts correctly with working shown. Interpretation of results is present but may be brief. At most one calculation error.	Attempts to work backwards but uses an incorrect method (e.g., divides instead of taking a root), OR solves only one part correctly. Working is incomplete. No meaningful interpretation of results is provided.
Final Synthesis — Answering the Driving Question	Directly and completely answers the driving question, addressing all four required points: why doubling gives ×8, the general rule for any k, how the pattern applies to design/build/predict, and a genuine personal reflection with a specific Kenyan example. Uses precise mathematical vocabulary throughout. The answer reads as a coherent, well-structured explanation.	Answers the driving question and addresses at least three of the four required points. Mathematical vocabulary is mostly correct. The reflection is present and connects to a Kenyan context, though it may be brief. The explanation is mostly coherent with minor gaps.	Attempts to answer the driving question but addresses fewer than three of the required points, OR the explanation of why doubling gives ×8 is incorrect or absent. Mathematical vocabulary is imprecise. The personal reflection is missing or does not connect to a Kenyan context.